

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 05



By- Satya sir

- Binary Heap

- children (or)
- Max-Heap min-heap

Parent at $\left\lfloor \frac{x-1}{2} \right\rfloor$

- Parent index, i $\rightarrow$ Left child index: $2i+1$
 $\rightarrow$ Right child index: $2i+2$



Topics to be Covered



- Deletion from Binary Heap

- Heap Orderings



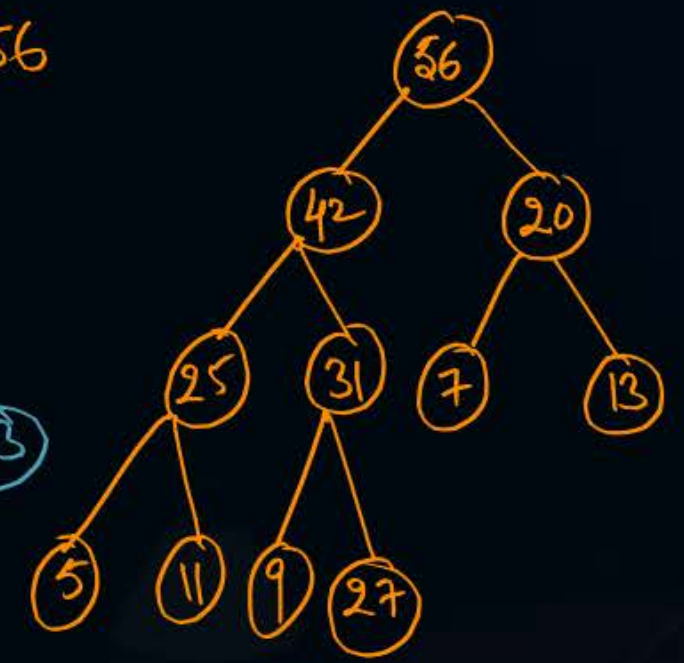
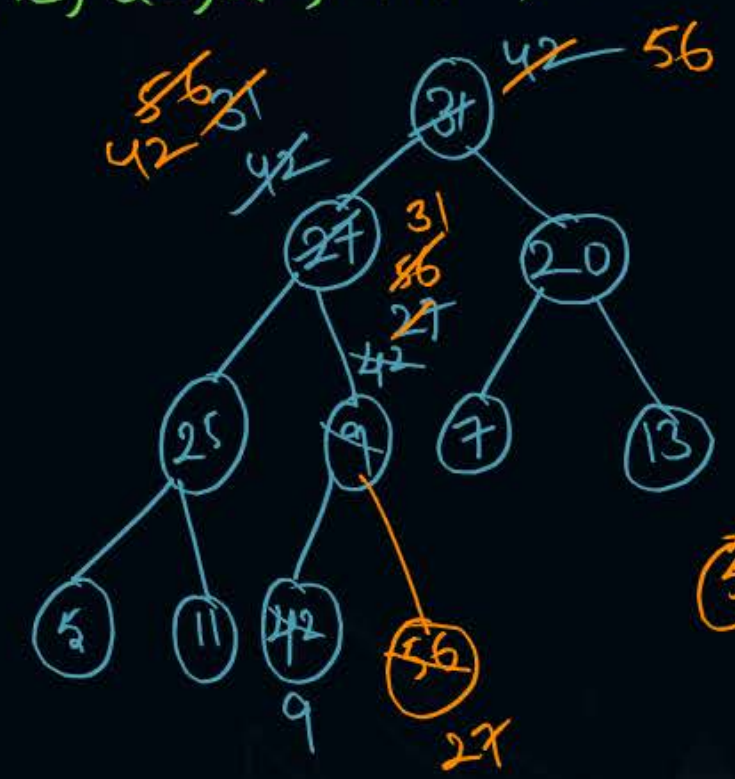
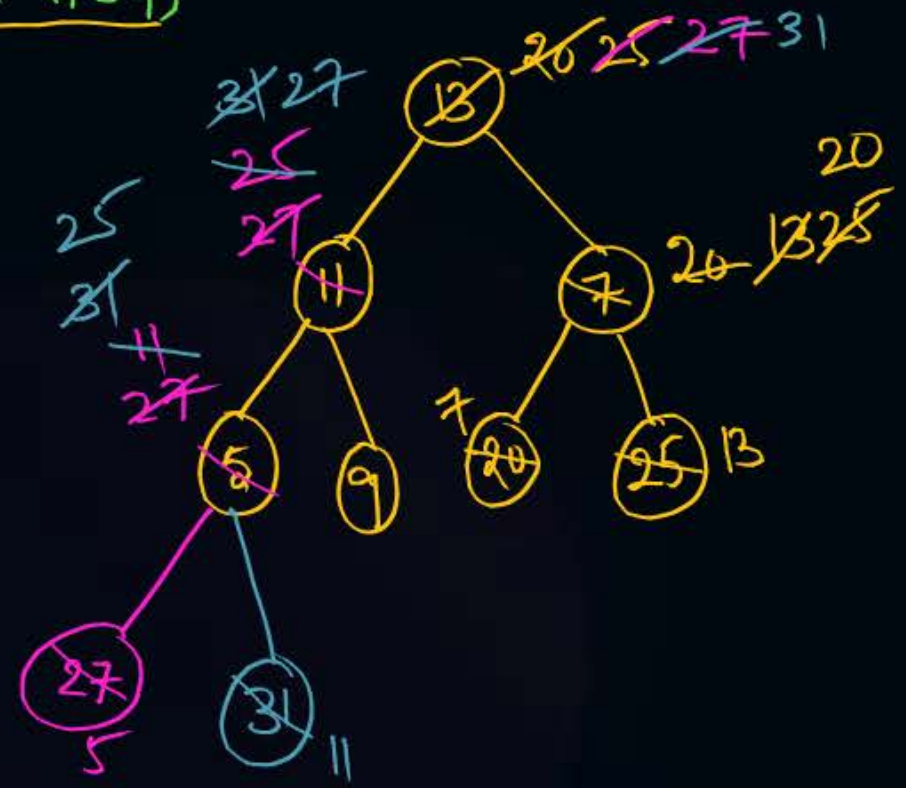
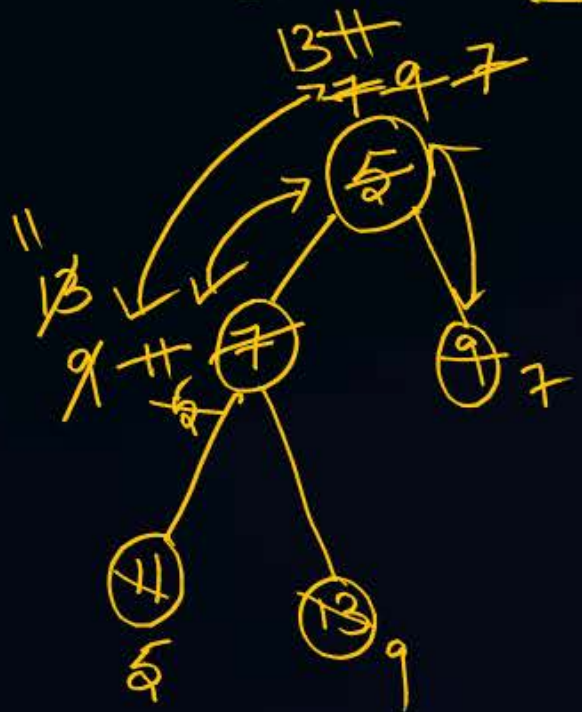


Topic : Binary Heap - 2



H/W

Construct Max-Heap, with Elements : 5, 7, 9, 11, 13, 20, 25, 27, 31, 42, 56



Resultant Max-Heap : 56, 42, 20, 25, 31, 7, 13, 5, 11, 9, 27



Topic : Binary Heap - 2



1) Deletion from Binary Heap : Deletion always happen at root node.

① Swap Root node with Last leaf node, Then Delete that leaf node.

② Heapify: $\left\{ \begin{array}{l} \rightarrow \text{For Max-heap: compare } [\text{Max}(\text{child}), \text{Parent}], \text{ if Parent} < \text{Max}(\text{child}), \text{ Then Swap.} \\ \rightarrow \text{For min-heap: compare } [\text{min}(\text{child}), \text{Parent}], \text{ if Parent} > \text{min}(\text{child}), \text{ Then Swap.} \end{array} \right.$

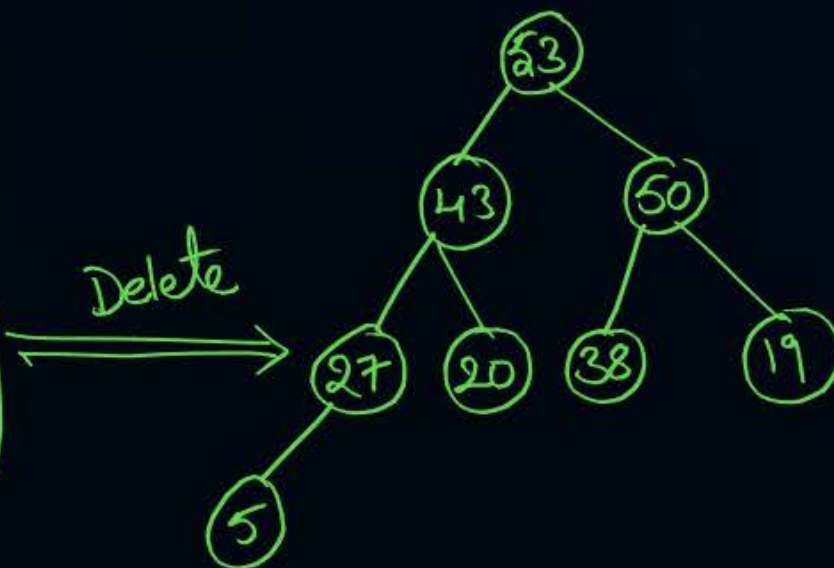
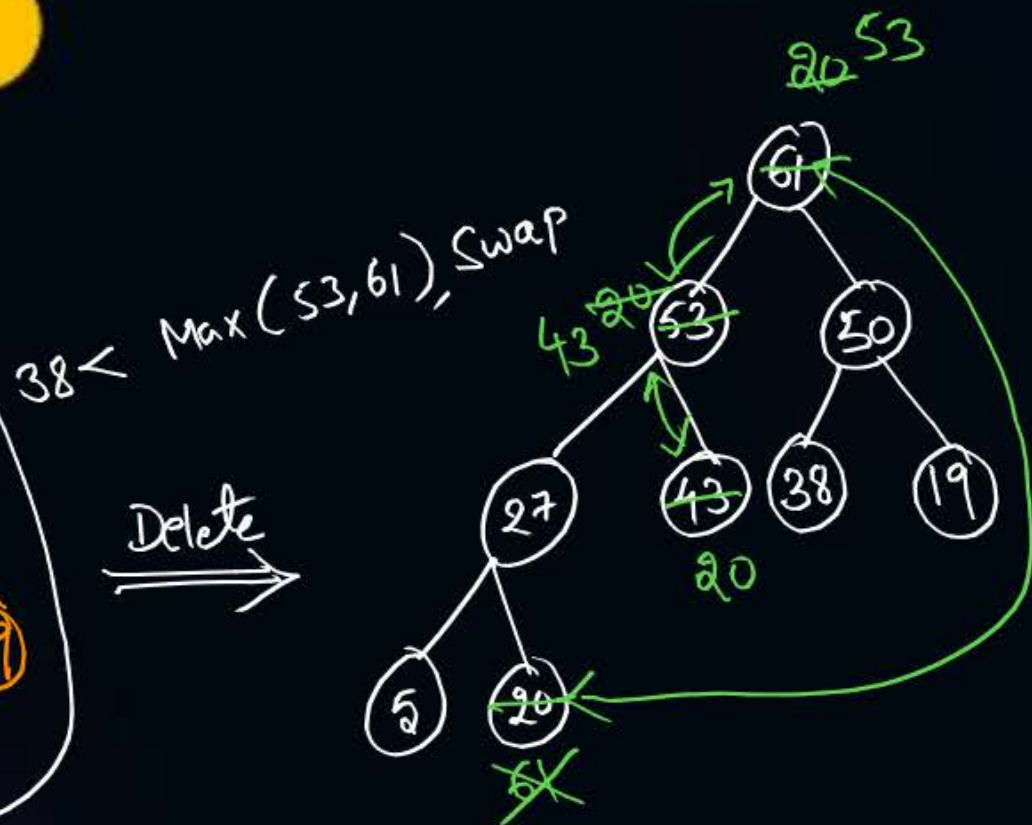
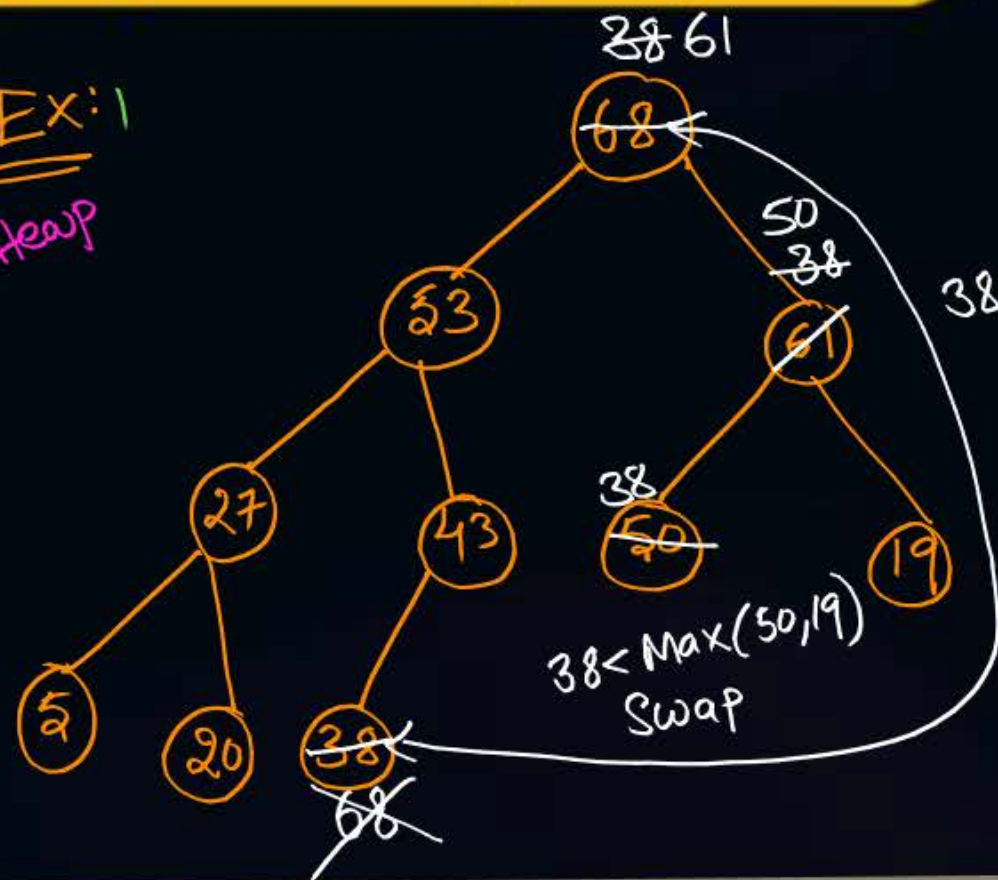
Repeat Heapification, until complete tree becomes a Binary Heap.



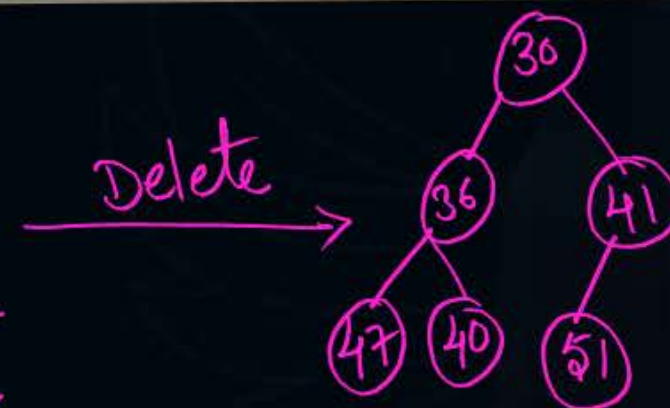
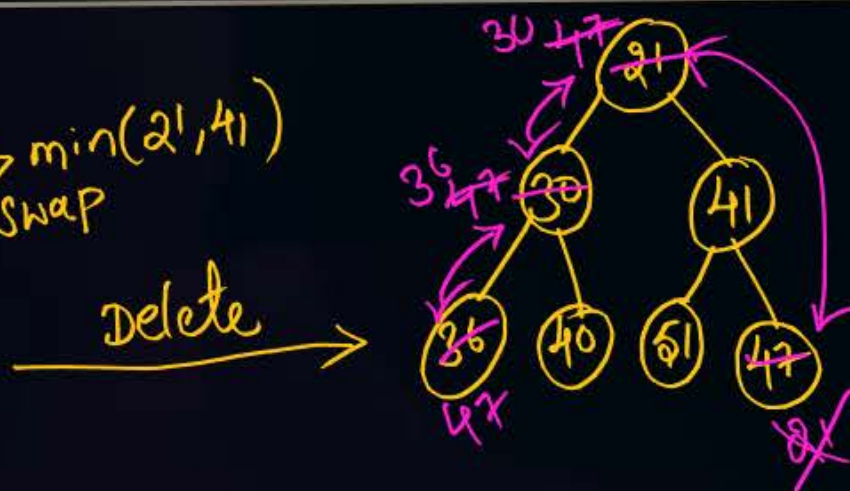
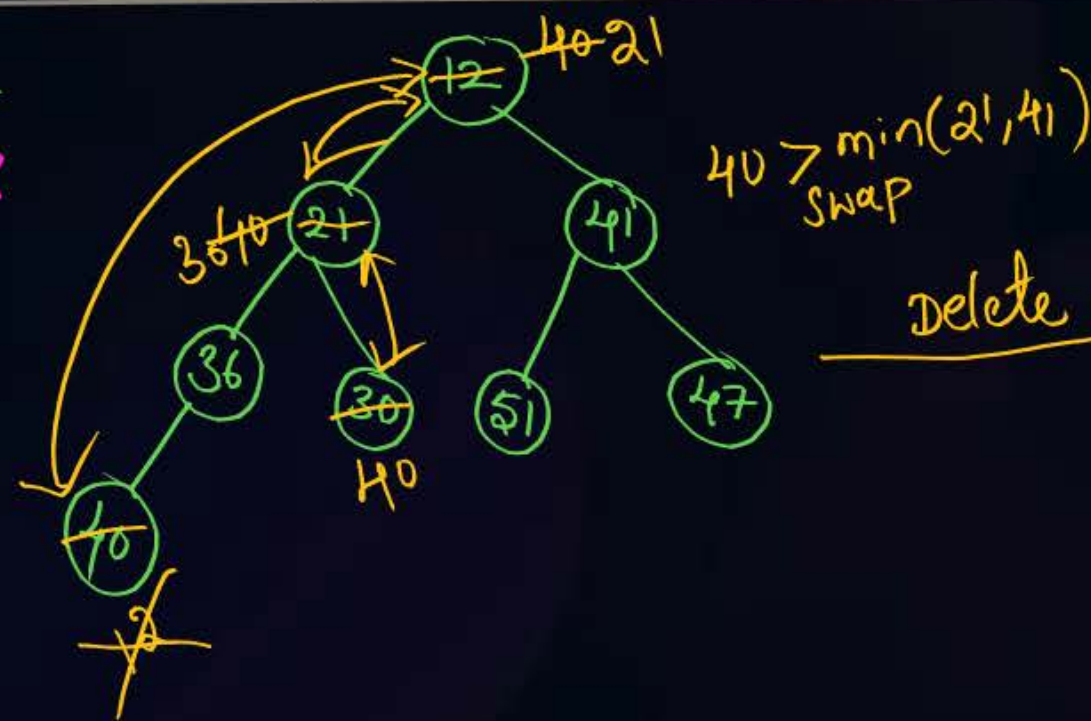
Topic : Binary Heap - 2



Ex:1
Max-Heap



Ex:2
min-heap





Topic : Binary Heap - 2



① Construct min-heap with elements 40, 5, 10, 7, 38, 21, 9, 3, 20, 11, 1

② Perform operations below in the same order, on above Constructed min-heap

a) Delete ✓

b) Insert 4 ✓

c) Insert 2 ✓

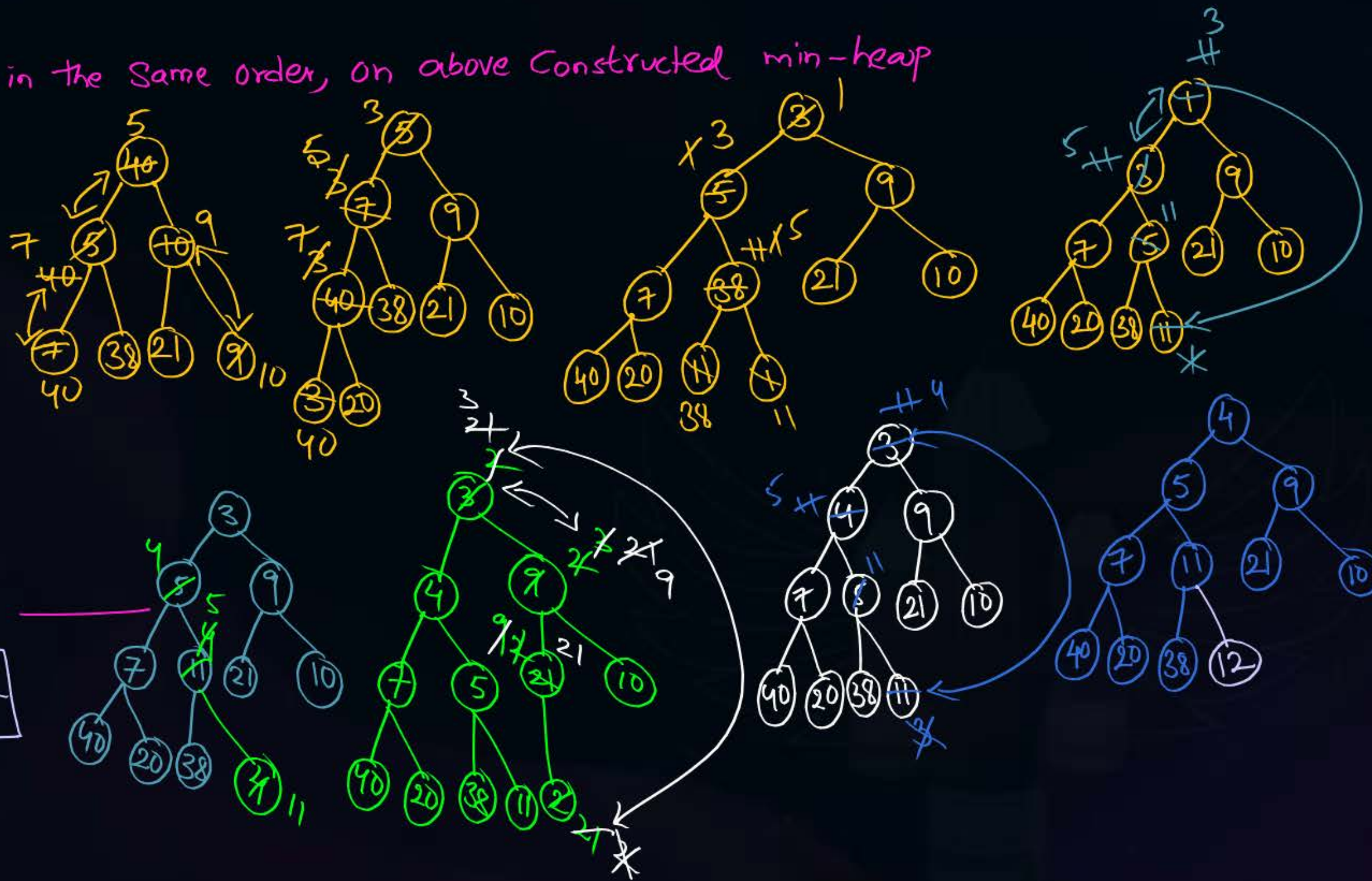
d) Delete ✓

e) Delete ✓

f) Insert 12 ✓

The Resultant min-heap is

4, 5, 9, 7, 11, 21, 10, 40, 20, 38, 12





Topic : Binary Heap - 2



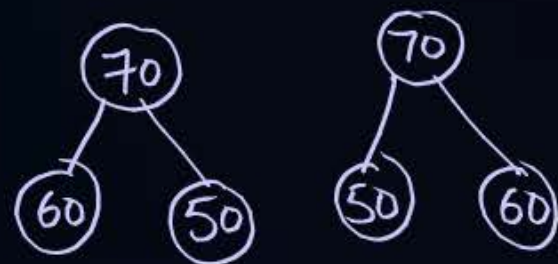
Heap orderings :

The Total number of Binary Heap [Max-heap/min-heap] that can be formed with 'N' elements will be

$$T(N) = {}^{(N-1)}_L C * T(L) * T(R)$$

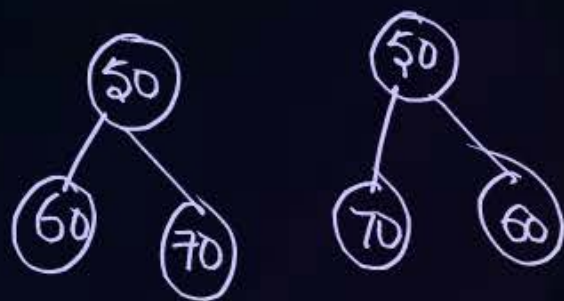
L = Left Subtree nodes
R = Right Subtree nodes

3 Elements: 70, 50, 60



Max-Heap

⇒ 2 binary Heap Possible.



min-heap

Ex: Let N = 7 elements

$$T(7) = {}^{(7-1)}_L C * T(L) * T(R)$$



$$\Rightarrow {}^6_3 C * T(3) * T(3)$$

$${}^6_3 C * \left[{}^{(3-1)}_1 C * T(1) * T(1) \right] * \left[{}^{(3-1)}_1 C * T(1) * T(1) \right]$$

$$= \frac{6*5*4*3!}{3!*3!} * (2*1*1) * (2*1*1) = \underline{\underline{80}}$$

$${}^n_r C = \frac{n!}{(n-r)! * r!}$$

⇒ Total Binary Heap orderings Possible with 7 elements = 80



Topic : Binary Heap - 2



Ex: $N=11$, find Total Binary heap orderings



$$T(11) = {}^{(11-1)}C_7 * T(7) * T(3)$$

$$= {}^{10}C_7 * \left[{}^6C_3 * T(3) * T(3) \right] * \left[{}^2C_1 * T(1) * T(1) \right]$$

$$= {}^{10}C_7 * \left[{}^6C_3 * 2 * 2 \right] * 2$$

$$= \frac{10 \times 9 \times 8 \times 7 \cancel{\times 6} \cancel{\times 5} \cancel{\times 4} \cancel{\times 3} \cancel{\times 2} \cancel{\times 1}}{\cancel{3} \cancel{\times 2} \cancel{\times 1}} * \frac{\cancel{6} \times 5 \times 4 \times \cancel{3} \cancel{\times 2} \cancel{\times 1}}{\cancel{3} \cancel{\times 2} \cancel{\times 1}} * 2 * 2 * 2 = 10 \times 12 \times 5 \times 4 \times 2 \times 2 \times 2 = \boxed{19200}$$



Topic : Binary Heap - 2



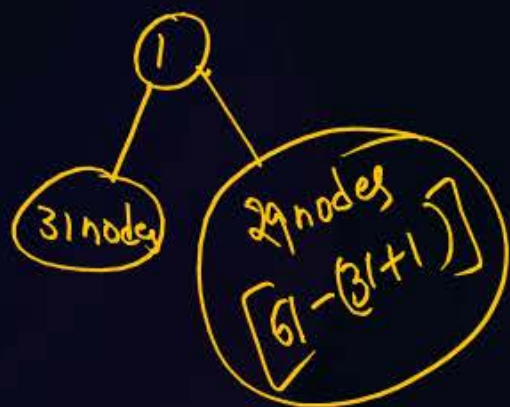
#Q. The Recurrence Relation for binary heap orderings with 61 elements, $T(61) =$ _____

$$T(61) = {}^{61}C_L * T(L) * T(R) \quad L = \underline{\hspace{2cm}} \quad R = \underline{\hspace{2cm}}$$

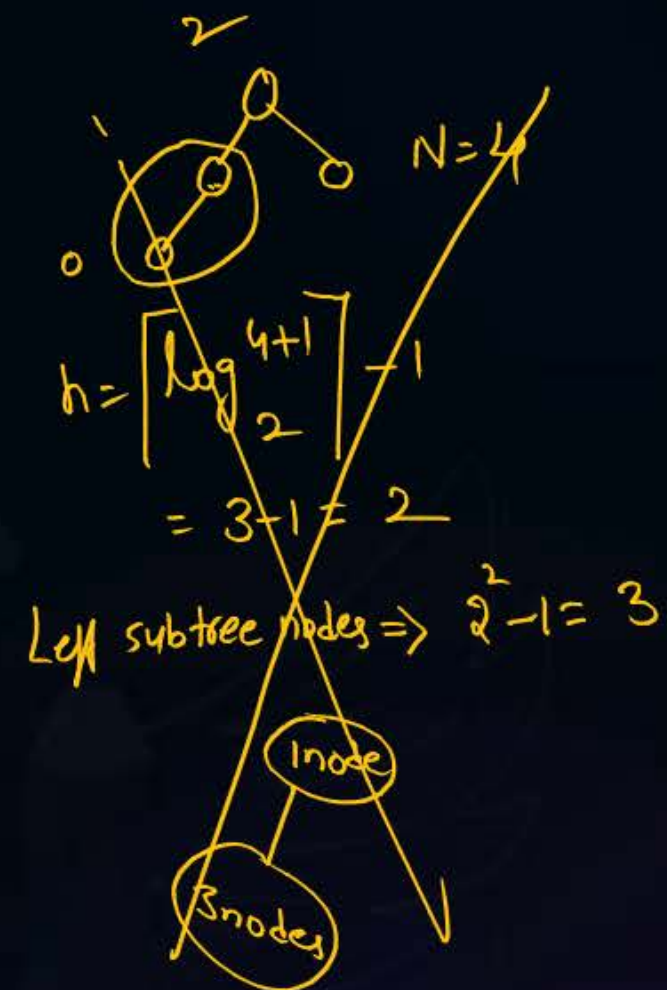
$$\Rightarrow \text{For CBT, Height, } h = \lceil \log_2^{N+1} \rceil - 1 \Rightarrow h = \lceil \log_2^{61+1} \rceil - 1 = 6 - 1 = \underline{5}$$

In CBT, Left subtree may be FBT, No. of nodes will be $2^h - 1$

$$\Rightarrow \text{Left subtree no. of nodes} = 2^5 - 1 = 32 - 1 = 31$$



$$\Rightarrow T(61) = {}^{61}C_{31} * T(31) * T(29)$$





Topic : Binary Heap - 2

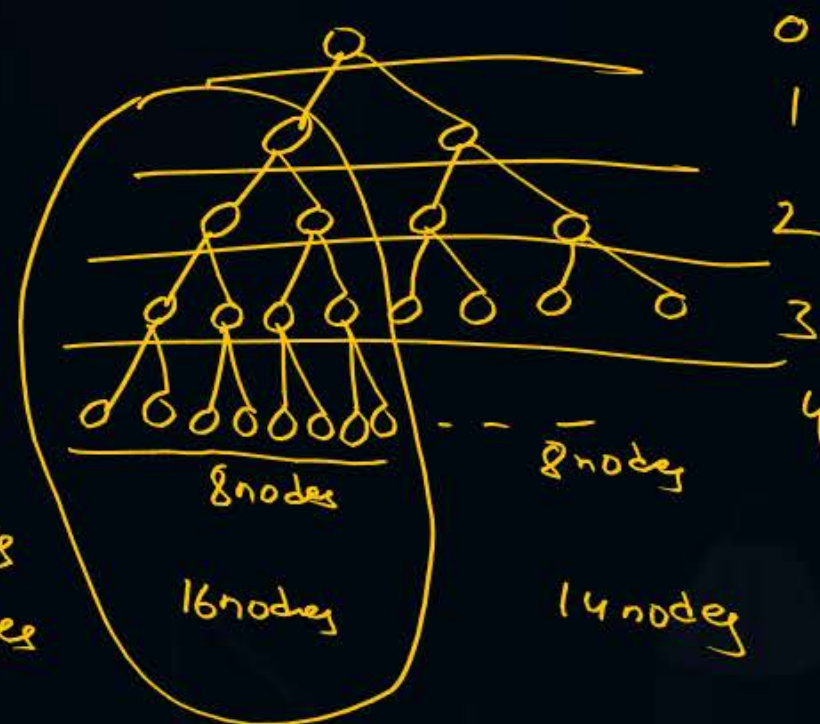


Total height / level of Tree = 5

At Level 0 = Root node = 1
Level 1 = 2^1 nodes = 2
Level 2 = 2^2 nodes = 4
Level 3 = 2^3 nodes = 8
Level 4 = 2^4 nodes = 16 — $\left\{ \begin{array}{l} \text{Left : 8 nodes} \\ \text{Right : 8 nodes} \end{array} \right.$
31 nodes

$$\text{Remaining} = 61 - 31 = 30$$

$$30 - 8 \times 2 = 14 \text{ Right}$$



Level / height = 2
4 nodes

Level $\Rightarrow$ 0 Root 1 node
Level $\Rightarrow$ 1 2^1 2 nodes — $\left\{ \begin{array}{l} \text{L} \\ \text{R} \end{array} \right.$
3 nodes

$$\text{Remaining} = 4 - 3 = 1 \text{ node}$$

Left nodes = 2 Right = 1



2 mins Summary



- Deletion from Binary heap
- Heap orderings

AVL Trees To be Contd...



THANK - YOU